

soccer_Solutions

Soccer, also known as football in many parts of the world, is a globally popular sport. Two teams of eleven players (4 defenders, 4 midfielders, 2 forwards, and 1 goalkeeper) compete to score goals by maneuvering a ball into the opposing team's net using any part of the body except their hands and arms (goalkeepers are the exception). The game emphasizes a blend of physical endurance, technical skill, and strategic teamwork. Games are 90 minutes long, divided into two 45-minute halves, with extra time and penalty shootouts used if necessary to determine a winner. Players are evaluated not only on goals and assists but also on their passing accuracy, defensive contributions, and overall impact on the game. Referees use yellow and red cards to manage player behavior and maintain fairness on the field. A yellow card serves as a warning for actions such as reckless tackles, delaying the game, or showing dissent toward the referee. If a player receives two yellow cards in one match, it results in a red card, meaning the player is ejected from the game. A red card can also be given directly for more serious offenses like violent conduct, serious foul play, or using offensive language. When a player is shown a red card, their team must continue the match with one fewer player.

The U.S. Men's National Soccer Team (USMNT) was officially established in 1913 while the U.S. Women's National Team (USWNT) was established in 1985. Players are selected for the U.S. national soccer teams through a combination of scouting, performance, and coaching decisions. For major tournaments like the World Cup or Olympics, coaches must select a limited number of players—usually around 23—who offer the best balance of skill, experience, and team dynamics. Veteran players must also go through the same process but do have a higher advantage than a new player.

Loading necessary packages:

```
library(readr)
library(dplyr)
library(ggplot2)
```

Reading in the Women's data set:

```
womensSoccer <- read_csv("womensSoccer2024.csv")
```

Reading in the Men's data set

```
mensSoccer <- read_csv("mensSoccer2024.csv")
```

Notice in both data sets the variable names are being treated as values instead of the column names.

- 1) Fix the `womensSoccer` and the `mensSoccer` data set so that the first row is the variable names.

```
colnames(womensSoccer) <- as.character(womensSoccer[1, ])
```

Error in eval(expr, envir, enclos): object 'womensSoccer' not found

```
womensSoccer1 <- womensSoccer[-1, ]
```

Error in eval(expr, envir, enclos): object 'womensSoccer' not found

```
colnames(mensSoccer) <- as.character(mensSoccer[1, ])
```

Error in eval(expr, envir, enclos): object 'mensSoccer' not found

```
mensSoccer1 <- mensSoccer[-1, ]
```

Error in eval(expr, envir, enclos): object 'mensSoccer' not found

- 2) Combine the men and women's data set into one data set.

```
soccer <- bind_rows(womensSoccer1, mensSoccer1)
```

Error in eval(expr, envir, enclos): object 'womensSoccer1' not found

a - What looks wrong in the new data set?

There are a bunch of NA values

b - Why do you think the data set doesn't look right?

The variable names are labeled differently in each data set and the men's soccer data set has added variables the women's data set does not have.

- 3) Change the variable names of the women's data set to match the men's and get rid of any variables in the men's that don't appear in the women's data set.

```
womenSoccerMatched <- womenSoccer1 |> rename(Player = Name, Pos = Pos., YC = Y, RC = R, `Car` = Car)
```

Error in eval(expr, envir, enclos): object 'womenSoccer1' not found

```
menSoccerMatched <- menSoccer1 |> select(1:7, 9:10, 12:13)
```

Error in eval(expr, envir, enclos): object 'menSoccer1' not found

Before we recombine the data sets, it would be helpful to know if the player is male or female and to exclude data not assigned to a player.

- 4) Add a new variable to both individual data sets that says "Male" or "Female" for gender of player.

```
womenSoccerMatched$Sex <- "Female"
```

Error: object 'womenSoccerMatched' not found

```
menSoccerMatched$Sex <- "Male"
```

Error: object 'menSoccerMatched' not found

- 5) Slice the Men's data set to exclude OWN GOAL, TOTALS, and OPPONENT TOTALS and slice the Women's data set to exclude Own Goal, TOTAL, and Opponent.

```
cleanedwomenSoccer <- womenSoccerMatched |> slice(1:39)
```

Error in eval(expr, envir, enclos): object 'womenSoccerMatched' not found

```
cleanedmenSoccer <- menSoccerMatched |> slice(1:48)
```

Error in eval(expr, envir, enclos): object 'menSoccerMatched' not found

- 6) Combine the the updated data sets.

```
combined_Soccer <- bind_rows(cleanedwomenSoccer, cleanedmenSoccer)
```

Error in eval(expr, envir, enclos): object 'cleanedwomenSoccer' not found

- 7) Change the variables to a numeric class except for Sex and Pos (change those to factor)

```
classSoccer <- combined_Soccer |> mutate(`Career Goals` = as.numeric(`Career Goals`), Pos = as.factor(Pos))
```

Error in eval(expr, envir, enclos): object 'combined_Soccer' not found

- 8) Create a scatterplot that uses Career Caps to predict Career Goals colored by Sex. Add a smoother for both genders and appropriate labels to the graph.

```
ggplot(data = classSoccer, aes(x = `Career Caps`, y = `Career Goals`, color = Sex)) +  
  geom_point() +  
  geom_smooth() +  
  labs(title = "Player's Career Caps vs. Career Goals",  
        x = "Total Minutes Played",  
        y = "Career Goals") +  
  theme_minimal()
```

Error in eval(expr, envir, enclos): object 'classSoccer' not found

- 9) Manipulate the dataset to get the highest scoring player for each gender.

```
topTwo <- classSoccer |>  
  group_by(Sex) |>  
  slice_max(order_by = `Career Goals`, n = 1) |>  
  ungroup()
```

Error in eval(expr, envir, enclos): object 'classSoccer' not found

- 10) Recreate the graph made in question 8 but add the player's name from question 9 to their respective point on the graph.

```
ggplot(data = classSoccer, aes(x = `Career Caps`, y = `Career Goals`, color = Sex)) +  
  geom_point() +  
  geom_smooth() +  
  geom_text(data = topTwo, aes(label = Player), vjust = -0.5, size = 3, show.legend = FALSE) +  
  labs(title = "Player's Career Caps vs. Career Goals",  
        x = "Career Caps",  
        y = "Career Goals") +  
  theme_minimal()
```

```
Error in eval(expr, envir, enclos): object 'classSoccer' not found
```